Signals & Systems (151-0575-01) Exam - February 8th, 2018

Combined question sheet and scoring rubric generated from JSON source. Criterion headings match rubric_manifest.json.

q1 - Consider the continuous-time system dq/dt(t) = [[0, 1], [0, 0]] q(t)... - 1 point:

Question (q1): Consider the continuous-time system dq/dt(t) = [[0, 1], [0, 0]] q(t) + [[1], [1]] u(t).

Question 1: What is the system description resulting from an exact discretization with sampling time Ts?
Options:
(1) q[n+1] = [[0, Ts], [0, 0]] q[n] + [[Ts], [Ts]] u[n]
(2) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[1], [1]] u[n]
(3) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts], [Ts]] u[n]
(4) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts (1 + Ts/2)], [Ts]] u[n]
Award full credit only for unambiguous selection of option (4): q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts (1 + Ts/2)], [Ts]] u[n]. Otherwise 0 unless the rubric allows partial credit.

q2 - Consider the continuous-time system dq/dt(t) = [[0, 1], [0, 0]] q(t)... - 1 point:

Question (q2): Consider the continuous-time system dq/dt(t) = [[0, 1], [0, 0]] q(t) + [[1], [1]] u(t).

Question 2: What is the system description resulting from a forward Euler discretization with sampling time Ts?
Options:
(1) q[n+1] = [[0, Ts], [0, 0]] q[n] + [[Ts], [Ts]] u[n]
(2) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[1], [1]] u[n]
(3) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts], [Ts]] u[n]
(4) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts (1 + Ts/2)], [Ts]] u[n]
Award full credit only for unambiguous selection of option (3): q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts], [Ts]] u[n]. Otherwise 0 unless the rubric allows partial credit.

q3 - Question 3: The continuous-time signal u(t) = cos(pi/10 t) is sampled... - 1 point:

Question (q3): Question 3: The continuous-time signal u(t) = cos(pi/10 t) is sampled with Ts = 0.1, resulting in a discrete-time sequence {u[n]}. The signal {u[n]} is recorded for N = 1000 consecutive time steps. How many periods of {u[n]} are recorded?
Options:
(1) 1
(2) 2
(3) 3
(4) 4
(5) 5
(6) 6
Award full credit only for unambiguous selection of option (5): 5. Otherwise 0 unless the rubric allows partial credit.

q4 - Consider a portion of the periodic continuous-time signal x(t), plott... - 1 point:

Question (q4): Consider a portion of the periodic continuous-time signal x(t), plotted for t from -4 to 8 (values roughly between -0.5 and 1.5). [Figure not included.]

Question 4: What is the fundamental period of the discrete-time signal {x[n]} if x(t) is sampled with Ts = 0.5?
Options:
(1) 1
(2) 2
(3) 4
(4) 6
(5) 8
Award full credit only for unambiguous selection of option (5): 8. Otherwise 0 unless the rubric allows partial credit.

q5 - Consider a portion of the periodic continuous-time signal x(t), plott... - 1 point:

Question (q5): Consider a portion of the periodic continuous-time signal x(t), plotted for t from -4 to 8 (values roughly between -0.5 and 1.5). [Figure not included.]

Question 5: What is the resulting discrete-time signal {x[n]} if x(t) is sampled with Ts = 1, where the first sample is taken at t = 0?
Options:
(1) x[n] = 0.5 + cos(pi/4 n) - 0.3 sin(3 pi n)
(2) x[n] = cos(pi/2 n)
(3) x[n] = 0.5 + sin(pi/4 n - pi/2)
(4) x[n] = 0.5 + cos(pi/2 n - pi/2)
(5) x[n] = sin(pi/2 n - pi/2)
(6) x[n] = cos(pi/4 n - pi/2) - 0.3 sin(3 pi n)
Award full credit only for unambiguous selection of option (4): x[n] = 0.5 + cos(pi/2 n - pi/2). Otherwise 0 unless the rubric allows partial credit.

q6 - Consider the continuous-time signal x(t) = cos(6pi/5 t) + cos(2 pi t)... - 1 point:

Question (q6): Consider the continuous-time signal x(t) = cos(6pi/5 t) + cos(2 pi t) and the discrete-time signals x_1[n] = 1 + cos(4pi/5 n), x_2[n] = 1 + cos(6pi/5 n), x_3[n] = cos(6pi/5 n + 3pi/10).

Question 6: Which discrete-time sequences result from sampling x(t) with sampling time Ts = 1, where the first sample is taken at t_0 = 0?
Options:
(1) None
(2) x_1[n]
(3) x_2[n]
(4) x_3[n]
(5) x_1[n] or x_2[n]
(6) x_1[n] or x_3[n]
(7) x_2[n] or x_3[n]
(8) x_1[n], x_2[n], or x_3[n]
Award full credit only for unambiguous selection of option (5): x_1[n] or x_2[n]. Otherwise 0 unless the rubric allows partial credit.

q7 - Consider the continuous-time signal x(t) = cos(6pi/5 t) + cos(2 pi t)... - 1 point:

Question (q7): Consider the continuous-time signal x(t) = cos(6pi/5 t) + cos(2 pi t) and the discrete-time signals x_1[n] = 1 + cos(4pi/5 n), x_2[n] = 1 + cos(6pi/5 n), x_3[n] = cos(6pi/5 n + 3pi/10).

Question 7: Which discrete-time sequences result from sampling x(t) with sampling time Ts = 1, where the first sample is taken at t_0 = 0.25?
Options:
(1) None
(2) x_1[n]
(3) x_2[n]
(4) x_3[n]
(5) x_1[n] or x_2[n]
(6) x_1[n] or x_3[n]
(7) x_2[n] or x_3[n]
(8) x_1[n], x_2[n], or x_3[n]
Award full credit only for unambiguous selection of option (4): x_3[n]. Otherwise 0 unless the rubric allows partial credit.

q8 - Consider the following systems: G_1: y[n] = u[|2n|] G_2: y[n] = u[n]... - 1 point:

Question (q8): Consider the following systems:
G_1: y[n] = u[|2n|]
G_2: y[n] = u[n] + u[n+1]
G_3: y[n] = 1 y[n-1] - 1/4 y[n-2] + u[n], causal
G_4: y[n] = 1/2 (y[n-1])^2 + u[n], causal

Question 8: Select all linear systems.
Options:
(1) G_1, G_2
(2) G_1, G_3

(3) G_1, G_4
(4) G_2, G_3
(5) G_2, G_4
(6) G_3, G_4
(7) G_1, G_2, G_3
(8) G_2, G_3, G_4
Award full credit only for unambiguous selection of option (7): G_1, G_2, G_3. Otherwise 0 unless the rubric allows partial credit.

q9 - Consider the following systems: G_1: y[n] = u[|2n|] G_2: y[n] = u[n]... - 1 point:

Question (q9): Consider the following systems:
G_1: y[n] = u[|2n|]
G_2: y[n] = u[n] + u[n+1]
G_3: y[n] = 1 y[n-1] - 1/4 y[n-2] + u[n], causal
G_4: y[n] = 1/2 (y[n-1])^2 + u[n], causal

Question 9: Select all time-invariant systems.
Options:
(1) G_1, G_2
(2) G_1, G_3
(3) G_1, G_4
(4) G_2, G_3
(5) G_2, G_4
(6) G_3, G_4
(7) G_1, G_2, G_3
(8) G_2, G_3, G_4
Award full credit only for unambiguous selection of option (8): G_2, G_3, G_4. Otherwise 0 unless the rubric allows partial credit.

q10 - Consider the following systems: G_1: y[n] = u[|2n|] G_2: y[n] = u[n]... - 1 point:

Question (q10): Consider the following systems:
G_1: y[n] = u[|2n|]
G_2: y[n] = u[n] + u[n+1]
G_3: y[n] = 1 y[n-1] - 1/4 y[n-2] + u[n], causal
G_4: y[n] = 1/2 (y[n-1])^2 + u[n], causal

Question 10: Select all bounded-input bounded-output stable systems.
Options:
(1) G_1, G_2
(2) G_1, G_3
(3) G_1, G_4
(4) G_2, G_3
(5) G_2, G_4
(6) G_3, G_4
(7) G_1, G_2, G_3
(8) G_2, G_3, G_4
Award full credit only for unambiguous selection of option (7): G_1, G_2, G_3. Otherwise 0 unless the rubric allows partial credit.

q11 - Consider the following representations of systems: G_1: h = {..., 0,... - 1 point:

Question (q11): Consider the following representations of systems:
G_1: h = {..., 0, 1 (n=0), 2, 3, 1, 0, ...}
G_2: y[n] = 2 y[n-1] + u[n], anti-causal
G_3: H(z) = z / (z^2 - 0.5 z + 0.25)

Question 11: Which systems are guaranteed to be stable?
Options:
(1) G_1
(2) G_2

(3) G_3
(4) G_1, G_2
(5) G_1, G_3
(6) G_2, G_3
(7) G_1, G_2, G_3
Award full credit only for unambiguous selection of option (4): G_1, G_2. Otherwise 0 unless the rubric allows partial credit.

q12 - Consider portions (n = -1..7) of the responses {y_1[n]} and {y_2[n]}... - 1 point:

Question (q12): Consider portions (n = -1..7) of the responses {y_1[n]} and {y_2[n]} of two systems to the unit step input {s[n]} (s[n] = 1 if n >= 0, 0 otherwise). [Figure not included.]

Question 12: Which of the system responses could result from an FIR filter?
Options:
(1) {y_1[n]}
(2) {y_2[n]}
(3) None of them
(4) {y_1[n]} and {y_2[n]}
Award full credit only for unambiguous selection of option (4): {y_1[n]} and {y_2[n]}. Otherwise 0 unless the rubric allows partial credit.

q13 - Consider portions (n = -1..7) of the responses {y_1[n]} and {y_2[n]}... - 1 point:

Question (q13): Consider portions (n = -1..7) of the responses {y_1[n]} and {y_2[n]} of two systems to the unit step input {s[n]} (s[n] = 1 if n >= 0, 0 otherwise). [Figure not included.]

Question 13: Consider the system response {y_1[n]}. Which difference equation could represent the system?
Options:
(1) y[n] = 1/2 y[n-1] + u[n]
(2) y[n] = 3/2 u[n] + u[n-1] + u[n-2]
(3) y[n] = -1/2 y[n-1] + 2 u[n]
(4) y[n] = 2 u[n] - 1/2 u[n-1]
(5) y[n] = 2/3 y[n-1] + u[n-1]
Award full credit only for unambiguous selection of option (5): y[n] = 2/3 y[n-1] + u[n-1]. Otherwise 0 unless the rubric allows partial credit.

q14 - Consider portions (n = -1..7) of the responses {y_1[n]} and {y_2[n]}... - 1 point:

Question (q14): Consider portions (n = -1..7) of the responses {y_1[n]} and {y_2[n]} of two systems to the unit step input {s[n]} (s[n] = 1 if n >= 0, 0 otherwise). [Figure not included.]

Question 14: Consider the system response {y_2[n]}. Which difference equation could represent the system?
Options:
(1) y[n] = 2/3 y[n] + u[n-1] + u[n-2]
(2) y[n] = 1/2 y[n-1] - 1/4 y[n-2] + u[n]
(3) y[n] = u[n] + 3/4 u[n-1] + 1/4 u[n-2]
(4) y[n] = u[n] + 1/2 u[n-1] + 1/2 u[n-2] + 1/2 u[n-3]
(5) y[n] = u[n] + 1/2 u[n-1] + 1/2 u[n-2]
Award full credit only for unambiguous selection of option (3): y[n] = u[n] + 3/4 u[n-1] + 1/4 u[n-2]. Otherwise 0 unless the rubric allows partial credit.

q15 - The impulse responses h_1, h_2, h_3, h_4 (n = 0..3) of four causal FI... - 1 point:

Question (q15): The impulse responses h_1, h_2, h_3, h_4 (n = 0..3) of four causal FIR filters are convolved with a sequence u (n = 0..50, values in {-1, 0, 1}), resulting in four output sequences y_A, y_B, y_C, y_D. [Figure not included.]

Question 15: Which output sequence y corresponds to y = h_3 * u?
Options:
(1) y = y_A
(2) y = y_B
(3) y = y_C
(4) y = y_D

Award full credit only for unambiguous selection of option (1): y = y_A. Otherwise 0 unless the rubric allows partial credit.

q16 - The impulse responses h_1, h_2, h_3, h_4 (n = 0..3) of four causal FI... - 1 point:

Question (q16): The impulse responses h_1, h_2, h_3, h_4 (n = 0..3) of four causal FIR filters are convolved with a sequence u (n = 0..50, values in {-1, 0, 1}), resulting in four output sequences y_A, y_B, y_C, y_D. [Figure not included.]

Question 16: Which impulse response h corresponds to y_B = h * u?
Options:
(1) h = h_1
(2) h = h_2
(3) h = h_3
(4) h = h_4
Award full credit only for unambiguous selection of option (4): h = h_4. Otherwise 0 unless the rubric allows partial credit.

q17 - Consider the following state-space descriptions of causal, LTI DT sys... - 1 point:

Question (q17): Consider the following state-space descriptions of causal, LTI DT systems, q[n+1] = A q[n] + B u[n], y[n] = C q[n] + D u[n]:
G_1: A = [[0, 1, 1], [0, 0, 1/2], [0, 0, 0]], B = [[1], [1], [-1]], C = [0, 0, 1], D = 0
G_2: A = 1, B = 1, C = 1, D = 0
G_3: A = [[0, 1/2], [1, 0]], B = [[1], [0]], C = [1, 1/2], D = 1
G_4: A = [[1/2, 0], [0, 0]], B = [[1], [1]], C = [1, 0], D = 0
G_5: A = 0, B = 1, C = 1, D = 1

Question 17: Select the answer containing all stable systems.
Options:
(1) G_1, G_2
(2) G_1, G_2, G_5
(3) G_5
(4) G_1, G_3, G_4, G_5
(5) G_1
(6) G_4
(7) G_1, G_5
(8) G_3
Award full credit only for unambiguous selection of option (4): G_1, G_3, G_4, G_5. Otherwise 0 unless the rubric allows partial credit.

q18 - Consider the following state-space descriptions of causal, LTI DT sys... - 1 point:

Question (q18): Consider the following state-space descriptions of causal, LTI DT systems, q[n+1] = A q[n] + B u[n], y[n] = C q[n] + D u[n]:
G_1: A = [[0, 1, 1], [0, 0, 1/2], [0, 0, 0]], B = [[1], [1], [-1]], C = [0, 0, 1], D = 0
G_2: A = 1, B = 1, C = 1, D = 0
G_3: A = [[0, 1/2], [1, 0]], B = [[1], [0]], C = [1, 1/2], D = 1
G_4: A = [[1/2, 0], [0, 0]], B = [[1], [1]], C = [1, 0], D = 0
G_5: A = 0, B = 1, C = 1, D = 1

Question 18: Select the answer containing all systems with an FIR.
Options:
(1) G_1
(2) G_2
(3) G_3
(4) G_4
(5) G_5
(6) G_1, G_2
(7) G_2, G_3
(8) G_4, G_5
(9) G_1, G_5
Award full credit only for unambiguous selection of option (9): G_1, G_5. Otherwise 0 unless the rubric allows partial credit.

q19 - Consider the following state-space descriptions of causal, LTI DT sys... - 1 point:

Question (q19): Consider the following state-space descriptions of causal, LTI DT systems, q[n+1] = A q[n] + B u[n], y[n] = C q[n] + D u[n]:
G_1: A = [[0, 1, 1], [0, 0, 1/2], [0, 0, 0]], B = [[1], [1], [-1]], C = [0, 0, 1], D = 0
G_2: A = 1, B = 1, C = 1, D = 0
G_3: A = [[0, 1/2], [1, 0]], B = [[1], [0]], C = [1, 1/2], D = 1
G_4: A = [[1/2, 0], [0, 0]], B = [[1], [1]], C = [1, 0], D = 0
G_5: A = 0, B = 1, C = 1, D = 1

Question 19: Which is a state-space representation of a system with LCCDE y[n] = 1/2 y[n-2] + u[n] + u[n-1]?
Options:
(1) G_1
(2) G_2
(3) G_3
(4) G_4
(5) G_5
Award full credit only for unambiguous selection of option (3): G_3. Otherwise 0 unless the rubric allows partial credit.

q20 - Consider portions (n = -4..4) of the impulse responses {h_i[n]} of st... - 1 point:

Question (q20): Consider portions (n = -4..4) of the impulse responses {h_i[n]} of stable LTI systems G_i, i in {1, ..., 6}. [Figure not included.]

Question 20: Which system has transfer function H(z) = z + (1/2 z) / (z - 1/2)?
Options:
(1) G_1
(2) G_2
(3) G_3
(4) G_4
(5) G_5
(6) G_6
Award full credit only for unambiguous selection of option (2): G_2. Otherwise 0 unless the rubric allows partial credit.

q21 - Consider a causal system with the difference equation y[n] = y[n-1] -... - 1 point:

Question (q21): Consider a causal system with the difference equation y[n] = y[n-1] - 1/4 y[n-2] + u[n-1].

Question 21: The input u[n] = 1 if n >= 0, 0 otherwise, is applied to the system. What is the value of y[n] for n -> infinity?
Options:
(1) 3
(2) 3.5
(3) 4
(4) 4.5
(5) 5
Award full credit only for unambiguous selection of option (3): 4. Otherwise 0 unless the rubric allows partial credit.

q22 - Consider a causal system with the difference equation y[n] = y[n-1] -... - 1 point:

Question (q22): Consider a causal system with the difference equation y[n] = y[n-1] - 1/4 y[n-2] + u[n-1].

Question 22: A periodic signal u[n] = 1/2 cos(pi n) is applied to the system. What is the output of the system?
Options:
(1) y[n] = 2 cos(pi n - pi)
(2) y[n] = 2/3 cos(pi n - pi)
(3) y[n] = 2/9 cos(pi n - pi)
(4) y[n] = 2 cos(pi n)
(5) y[n] = 2/3 cos(pi n)
(6) y[n] = 2/9 cos(pi n)
Award full credit only for unambiguous selection of option (3): y[n] = 2/9 cos(pi n - pi). Otherwise 0 unless the rubric allows partial credit.

q23 - Consider a causal LTI system G_* (y_* = G_* u_*) given by its magnitu... - 1 point:

Question (q23): Consider a causal LTI system G_* (y_* = G_* u_*) given by its magnitude and phase frequency response plots for Omega in [0, pi]. [Figure not included.] G_* is used to create two systems: G_A is the series connection G_* -> G_*; G_B is the parallel connection of two copies of G_*, whose outputs are summed. The input to both G_A and G_B is u[n] = 2 cos(pi/2 n) + 3/2 cos(2pi/3 n) + sin(pi n).

Question 23: What is the resulting output y = G_A u?
Options:
(1) y[n] = 2/9 cos(pi/2 n) + 1/9 sin(pi n)
(2) y[n] = -4/3 cos(pi/2 n) + cos(2pi/3 n) + 2/3 sin(pi n)
(3) y[n] = 4/3 sin(pi/2 n - pi/2) + 2/3 sin(pi n)
(4) y[n] = 2/3 cos(pi/2 n) + 3/2 cos(2pi/3 n) + 1/3 sin(pi n)
(5) y[n] = -2/9 cos(pi/2 n) + 1/9 sin(pi n)
(6) y[n] = 2/3 sin(pi/2 n) + 1/3 sin(pi n)
Award full credit only for unambiguous selection of option (5): y[n] = -2/9 cos(pi/2 n) + 1/9 sin(pi n). Otherwise 0 unless the rubric allows partial credit.

q24 - Consider a causal LTI system G_* (y_* = G_* u_*) given by its magnitu... - 1 point:

Question (q24): Consider a causal LTI system G_* (y_* = G_* u_*) given by its magnitude and phase frequency response plots for Omega in [0, pi]. [Figure not included.] G_* is used to create two systems: G_A is the series connection G_* -> G_*; G_B is the parallel connection of two copies of G_*, whose outputs are summed. The input to both G_A and G_B is u[n] = 2 cos(pi/2 n) + 3/2 cos(2pi/3 n) + sin(pi n).

Question 24: What is the resulting output y = G_B u?
Options:
(1) y[n] = -4/3 cos(pi/2 n) + cos(2pi/3 n) + 2/3 sin(pi n)
(2) y[n] = 2/3 sin(pi/2 n) + 1/3 sin(pi n)
(3) y[n] = -2/9 cos(pi/2 n) + 1/9 sin(pi n)
(4) y[n] = 4/3 cos(pi/2 n - pi/2) + 2/3 sin(pi n)
(5) y[n] = 2/9 cos(pi/2 n) + 1/9 sin(pi n)
(6) y[n] = 2/3 cos(pi/2 n) + 3/2 cos(2pi/3 n) + 1/3 sin(pi n)
Award full credit only for unambiguous selection of option (4): y[n] = 4/3 cos(pi/2 n - pi/2) + 2/3 sin(pi n). Otherwise 0 unless the rubric allows partial credit.

q25 - Consider DT Fourier magnitude plots of four different causal sequence... - 1 point:

Question (q25): Consider DT Fourier magnitude plots of four different causal sequences: X_A(Omega), X_B(Omega), X_C(Omega), X_D(Omega), for Omega in [0, pi]. [Figure not included.]

Question 25: Which could be the magnitude plot corresponding to a given causal sequence x[n] (stem plot for n = 0..10)? [Figure not included.]
Options:
(1) X_A(Omega)
(2) X_B(Omega)
(3) X_C(Omega)
(4) X_D(Omega)
Award full credit only for unambiguous selection of option (1): X_A(Omega). Otherwise 0 unless the rubric allows partial credit.

q26 - Consider DT Fourier magnitude plots of four different causal sequence... - 1 point:

Question (q26): Consider DT Fourier magnitude plots of four different causal sequences: X_A(Omega), X_B(Omega), X_C(Omega), X_D(Omega), for Omega in [0, pi]. [Figure not included.]

Question 26: Which could be the magnitude plot corresponding to another given causal sequence x[n] (stem plot for n = 0..10)? [Figure not included.]
Options:
(1) X_A(Omega)
(2) X_B(Omega)
(3) X_C(Omega)
(4) X_D(Omega)
Award full credit only for unambiguous selection of option (2): X_B(Omega). Otherwise 0 unless the rubric allows partial credit.

q27 - Consider the magnitude and phase response of a stable LTI system G, f... - 1 point:

Question (q27): Consider the magnitude and phase response of a stable LTI system G, for Omega in [0, pi]. [Figure not included.]

Question 27: Which is the transfer function of the LTI system G?
Options:
(1) H(z) = 0.5 (z + 1) / (z - 0.5)
(2) H(z) = 0.35 (z + 1) / (z - 0.3)
(3) H(z) = (z + 1)^2 / (4 z^2)
(4) H(z) = (z + 1) / (2z)
(5) H(z) = (z + 0.73) / (z^2 - 0.3)
Award full credit only for unambiguous selection of option (2): H(z) = 0.35 (z + 1) / (z - 0.3). Otherwise 0 unless the rubric allows partial credit.

q28 - Consider the magnitude and phase response of a stable LTI system G, f... - 1 point:

Question (q28): Consider the magnitude and phase response of a stable LTI system G, for Omega in [0, pi]. [Figure not included.]

Question 28: The LTI system is cascaded with a delay element G_D with transfer function H_D(z) = z^-1 (u -> G -> G_D -> y). Which of four magnitude/phase plots (Plot A to Plot D) shows the response of the resulting system G_casc? [Figure not included.]
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
Award full credit only for unambiguous selection of option (4): Plot D. Otherwise 0 unless the rubric allows partial credit.

q29 - Question 29: The sequence {x[n]} = k * {-1 (n=0), -2, -1, 1, 2, 1} is... - 1 point:

Question (q29): Question 29: The sequence {x[n]} = k * {-1 (n=0), -2, -1, 1, 2, 1} is periodic with period N = 6, where k is an integer. The corresponding DFS coefficients are {X[k]} = {0 (k=0), X[1], 0, 0, 0, X[5]}. What is the magnitude of coefficient X[1]?
Options:
(1) |X[1]| = 3|k|^2
(2) |X[1]| = sqrt(72)|k|
(3) |X[1]| = 6|k|
(4) |X[1]| = 36|k|
(5) |X[1]| = 36|k|^2
(6) |X[1]| = 3|k|
Award full credit only for unambiguous selection of option (3): |X[1]| = 6|k|. Otherwise 0 unless the rubric allows partial credit.

q30 - Consider the DT signal x[n] = 2 e^(j 4pi/7 n) + 2 e^(j 10pi/7 n) for... - 1 point:

Question (q30): Consider the DT signal x[n] = 2 e^(j 4pi/7 n) + 2 e^(j 10pi/7 n) for all times n. A DFT of the first N = 21 samples of {x[n]} is taken. Let {X[k]} be the corresponding DFT coefficients.

Question 30: What is the value of X[1]?
Options:
(1) X[1] = 2/3
(2) X[1] = 1/3
(3) X[1] = 42
(4) X[1] = 2
(5) X[1] = 6
(6) X[1] = 0
Award full credit only for unambiguous selection of option (6): X[1] = 0. Otherwise 0 unless the rubric allows partial credit.

q31 - Consider the DT signal x[n] = 2 e^(j 4pi/7 n) + 2 e^(j 10pi/7 n) for... - 1 point:

Question (q31): Consider the DT signal x[n] = 2 e^(j 4pi/7 n) + 2 e^(j 10pi/7 n) for all times n. A DFT of the first N = 21 samples of {x[n]} is taken. Let {X[k]} be the corresponding DFT coefficients.

Question 31: What is the value of X[6]?
Options:
(1) X[6] = 2
(2) X[6] = 6
(3) X[6] = 42
(4) X[6] = 0
(5) X[6] = 2/3
(6) X[6] = 1/3
Award full credit only for unambiguous selection of option (3): X[6] = 42. Otherwise 0 unless the rubric allows partial credit.

q32 - Question 32: The magnitudes of the DFT coefficients computed with the... - 1 point:

Question (q32): Question 32: The magnitudes of the DFT coefficients computed with the first N = 4 samples of a real discrete-time signal {x[n]} are |X[k]| = {0, 2, 0, 2} for k = 0..3. Which of four plots (Plot A to Plot D) could show the magnitudes of the DFT coefficients computed with the first N = 6 samples of the same signal? [Figure not included.]
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
Award full credit only for unambiguous selection of option (3): Plot C. Otherwise 0 unless the rubric allows partial credit.

q33 - Question 33: Consider the transfer function of system G: H(z) = (z^2... - 1 point:

Question (q33): Question 33: Consider the transfer function of system G: H(z) = (z^2 + 1/2 z) / (z^2 - 1/4). What is the impulse response of G?
Options:
(1) h = {..., 0, 2 (n=0), 1, 1, 1, 0, ...}
(2) h = {..., 0, 2 (n=0), 1, 1/2, 1/4, 0, ...}
(3) h = {..., 0, 1 (n=0), 1/3, 1/9, 1/27, ...}
(4) h = {..., 0, 1 (n=0), 1/2, 1/4, 1/8, ...}
(5) h = {..., 0, 1, 1 (n=0), 1/2, 1/4, 0, ...}
(6) h = {..., 3, 1 (n=0), 1/3, 1/9, 1/27, ...}
Award full credit only for unambiguous selection of option (4): h = {..., 0, 1 (n=0), 1/2, 1/4, 1/8, ...}. Otherwise 0 unless the rubric allows partial credit.

q34 - Question 34: Which of six magnitude responses (Plot A to Plot F) belo... - 1 point:

Question (q34): Question 34: Which of six magnitude responses (Plot A to Plot F) belongs to a system with transfer function H(z) = 1/8 (z - 1) / (z^2 + z + 0.5)? [Figure not included.]
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
(5) Plot E
(6) Plot F
Award full credit only for unambiguous selection of option (1): Plot A. Otherwise 0 unless the rubric allows partial credit.

q35 - Question 35: Let G be a causal LTI system whose pole-zero plot is giv... - 1 point:

Question (q35): Question 35: Let G be a causal LTI system whose pole-zero plot is given (poles x, zeros o, unit circle shown). [Figure not included.] Which of four magnitude responses (Plot A to Plot D) could belong to system G? [Figure not included.]
Options:
(1) Plot A
(2) Plot B
(3) Plot C
(4) Plot D
Award full credit only for unambiguous selection of option (1): Plot A. Otherwise 0 unless the rubric allows partial credit.

q36 - Question 36: Consider a CT low-pass filter H_LP(s) and a CT high-pass... - 1 point:

Question (q36): Question 36: Consider a CT low-pass filter H_LP(s) and a CT high-pass filter H_HP(s), given by their Bode magnitude plots for omega from 10^-3 to 10^3 rad/s. [Figure not included.] Which frequency transformation was used to obtain H_HP(s) from H_LP(s)?
Options:
(1) H_HP(s) = H_LP(1/(10s))
(2) H_HP(s) = H_LP(s^2)
(3) H_HP(s) = H_LP((s^2 + 1)/s)
(4) H_HP(s) = H_LP(s/10)
(5) H_HP(s) = H_LP(10/s)
(6) H_HP(s) = H_LP(10s)
Award full credit only for unambiguous selection of option (1): H_HP(s) = H_LP(1/(10s)). Otherwise 0 unless the rubric allows partial credit.

q37 - Consider the system G built as follows: the input u goes through G_1... - 1 point:

Question (q37): Consider the system G built as follows: the input u goes through G_1 followed by G_2 (series path), and in parallel through G_3; the outputs of the two paths are summed to give y. Available systems for G_i are gains H_i(z) = a, low-pass filters H_i(z) = H_LP(z), and high-pass filters H_i(z) = H_HP(z). Note that you only have to choose the class of system, not its specifications.

Question 37: How can G_1, G_2 and G_3 be chosen to design G to be a band-stop filter?
Options:
(1) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 0
(2) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_LP(z)
(3) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_HP(z)
(4) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 1
(5) G_1: H_1(z) = H_HP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_LP(z)
(6) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_HP(z)
Award full credit only for unambiguous selection of option (3): G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_HP(z). Otherwise 0 unless the rubric allows partial credit.

q38 - Consider the system G built as follows: the input u goes through G_1... - 1 point:

Question (q38): Consider the system G built as follows: the input u goes through G_1 followed by G_2 (series path), and in parallel through G_3; the outputs of the two paths are summed to give y. Available systems for G_i are gains H_i(z) = a, low-pass filters H_i(z) = H_LP(z), and high-pass filters H_i(z) = H_HP(z). Note that you only have to choose the class of system, not its specifications.

Question 38: How can G_1, G_2 and G_3 be chosen to design G to be a band-pass filter?
Options:
(1) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_HP(z)
(2) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_HP(z)
(3) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 1
(4) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_LP(z)
(5) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 0
(6) G_1: H_1(z) = H_HP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_LP(z)
Award full credit only for unambiguous selection of option (5): G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 0. Otherwise 0 unless the rubric allows partial credit.

q39 - A second-order IIR filter with transfer function H(z) = (b_0 + b_1 z^... - 1 point:

Question (q39): A second-order IIR filter with transfer function H(z) = (b_0 + b_1 z^-1 + b_2 z^-2) / (1 + a_1 z^-1 + a_2 z^-2) (1) is realized with the following block diagram. On the left, u passes through two cascaded delay elements G_d, giving u[n-1] and u[n-2]; u[n], u[n-1], u[n-2] are scaled by gains g_0, g_1, g_2. On the right, y passes through two cascaded delay elements G_d, giving y[n-1] and y[n-2], scaled by gains g_4 and g_5. A bottom adder sums g_2 u[n-2] and g_5 y[n-2]; a middle adder sums g_1 u[n-1], g_4 y[n-1] and the bottom adder output; a top adder sums g_0 u[n] and the middle adder output. The top adder output is scaled by g_3 to give y. The delay elements have transfer function H_d(z) = z^-1, and a gain g maps x_1 to x_2 = g x_1, g in R.

Question 39: How should the gains be chosen such that the second-order IIR filter (1) is realized?
Options:
(1) g_0 = 1, g_1 = a_1, g_2 = a_2, g_3 = b_0, g_4 = b_1, g_5 = b_2
(2) g_0 = 1, g_1 = a_1, g_2 = a_2, g_3 = b_0, g_4 = -b_1, g_5 = -b_2

(3) g_0 = b_0, g_1 = b_1, g_2 = b_2, g_3 = 1, g_4 = a_1, g_5 = a_2
(4) g_0 = b_0, g_1 = b_1, g_2 = b_2, g_3 = 1, g_4 = -a_1, g_5 = -a_2
Award full credit only for unambiguous selection of option (4): g_0 = b_0, g_1 = b_1, g_2 = b_2, g_3 = 1, g_4 = -a_1, g_5 = -a_2. Otherwise 0 unless the rubric allows partial credit.

q40 - A second-order IIR filter with transfer function H(z) = (b_0 + b_1 z^... - 1 point:

Question (q40): A second-order IIR filter with transfer function H(z) = (b_0 + b_1 z^-1 + b_2 z^-2) / (1 + a_1 z^-1 + a_2 z^-2) (1) is realized with the following block diagram. On the left, u passes through two cascaded delay elements G_d, giving u[n-1] and u[n-2]; u[n], u[n-1], u[n-2] are scaled by gains g_0, g_1, g_2. On the right, y passes through two cascaded delay elements G_d, giving y[n-1] and y[n-2], scaled by gains g_4 and g_5. A bottom adder sums g_2 u[n-2] and g_5 y[n-2]; a middle adder sums g_1 u[n-1], g_4 y[n-1] and the bottom adder output; a top adder sums g_0 u[n] and the middle adder output. The top adder output is scaled by g_3 to give y. The delay elements have transfer function H_d(z) = z^-1, and a gain g maps x_1 to x_2 = g x_1, g in R.

Question 40: The form above needs four delay units to realize a second-order IIR filter with transfer function (1). Which statement is correct?
Options:
(1) Five is the minimum number of delay elements needed to realize a general second order IIR filter. Four elements were only sufficient above because of a_0 = 1 in (1).
(2) Four is the minimum number of delay elements needed to realize a general second order IIR filter.
(3) Three is the minimum number of delay elements needed to realize a general second order IIR filter.
(4) Two is the minimum number of delay elements needed to realize a general second order IIR filter.
(5) One is the minimum number of delay elements needed to realize a general second order IIR filter.
Award full credit only for unambiguous selection of option (4): Two is the minimum number of delay elements needed to realize a general second order IIR filter.. Otherwise 0 unless the rubric allows partial credit.

q41 - Consider a sequence {u[n]}, n = 0..100, filtered by four different ca... - 1 point:

Question (q41): Consider a sequence {u[n]}, n = 0..100, filtered by four different causal discrete-time filters G_i, i in {1, 2, 3, 4}, resulting in the sequences y_i = G_i u. [Figure not included.]

Question 41: Which filter could have the given magnitude response (close to 0 at Omega = 0, rising to about 1 by Omega = pi/4 and staying near 1 up to pi)?
Options:
(1) G_2
(2) G_4
(3) G_1
(4) G_3
Award full credit only for unambiguous selection of option (2): G_4. Otherwise 0 unless the rubric allows partial credit.

q42 - Consider a sequence {u[n]}, n = 0..100, filtered by four different ca... - 1 point:

Question (q42): Consider a sequence {u[n]}, n = 0..100, filtered by four different causal discrete-time filters G_i, i in {1, 2, 3, 4}, resulting in the sequences y_i = G_i u. [Figure not included.]

Question 42: Which could be the gain of the filter G_3 at Omega = pi/2?
Options:
(1) |H_3(Omega = pi/2)| = 0
(2) |H_3(Omega = pi/2)| = 2
(3) |H_3(Omega = pi/2)| = 4
(4) |H_3(Omega = pi/2)| = 0.98
(5) |H_3(Omega = pi/2)| = pi
Award full credit only for unambiguous selection of option (1): |H_3(Omega = pi/2)| = 0. Otherwise 0 unless the rubric allows partial credit.

q43 - Consider a sequence {u[n]}, n = 0..100, filtered by four different ca... - 1 point:

Question (q43): Consider a sequence {u[n]}, n = 0..100, filtered by four different causal discrete-time filters G_i, i in {1, 2, 3, 4}, resulting in the sequences y_i = G_i u. [Figure not included.]

Question 43: Select the answer containing all filter(s) which could have an FIR shorter than 4 samples.

Options:
(1) G_1
(2) G_2
(3) G_3
(4) G_4
(5) G_1, G_2
(6) G_1, G_3
(7) G_1, G_4
(8) G_2, G_3
(9) G_2, G_4
Award full credit only for unambiguous selection of option (6): G_1, G_3. Otherwise 0 unless the rubric allows partial credit.

q44 - Consider a sequence {u[n]}, n = 0..100, filtered by four different ca... - 1 point:

Question (q44): Consider a sequence {u[n]}, n = 0..100, filtered by four different causal discrete-time filters G_i, i in {1, 2, 3, 4}, resulting in the sequences y_i = G_i u. [Figure not included.]

Question 44: Which filter could be a median filter?
Options:
(1) G_1
(2) G_2
(3) G_3
(4) G_4
Award full credit only for unambiguous selection of option (1): G_1. Otherwise 0 unless the rubric allows partial credit.

q45 - Question 45: Consider a continuous-time, causal system with transfer... - 1 point:

Question (q45): Question 45: Consider a continuous-time, causal system with transfer function H(s) = (beta s + 1) / (alpha s^2 + s + 1) with alpha > 0 and beta > 0. The system is discretized using the bilinear transform and sampling time Ts. What is the phase of the discrete-time transfer function, angle H(z), at z = e^(j pi)?
Options:
(1) angle H(z) = -pi/2
(2) angle H(z) = -pi/4
(3) angle H(z) = 0
(4) angle H(z) = pi/2
(5) angle H(z) = pi/4
(6) angle H(z) = pi
Award full credit only for unambiguous selection of option (1): angle H(z) = -pi/2. Otherwise 0 unless the rubric allows partial credit.

q46 - Consider the following system descriptions: G_1: h = {..., 0, 1 (n=0)... - 1 point:

Question (q46): Consider the following system descriptions:
G_1: h = {..., 0, 1 (n=0), 1/2, 1/3, 0, ...}
G_2: y[n] = 1/2 y[n-1] + u[n]
G_3: y[n] = u[n-1]
G_4: y[n] = u[n+1] + 2 u[n] + u[n-1]

Question 46: Select all systems whose phase response is zero.
Options:
(1) G_1
(2) G_2
(3) G_3
(4) G_4
(5) G_1, G_2
(6) G_1, G_3
(7) G_1, G_4
(8) G_2, G_3
(9) G_2, G_4
(10) G_3, G_4
Award full credit only for unambiguous selection of option (4): G_4. Otherwise 0 unless the rubric allows partial credit.

q47 - The objective of the following MATLAB code is to estimate the transfe... - 1 point:

Question (q47): The objective of the following MATLAB code is to estimate the transfer function coefficients a_1, b_0 and b_1 of H(z) = (b_0 + b_1 z^-1) / (1 + a_1 z^-1), given estimated frequency responses H_hat(Omega) for frequencies INCLUDING Omega = 0 and Omega = pi, saved in MATLAB vectors H.mat and Omega.mat, respectively.

```
1 %% First order transfer function estimation
2
3 % load estimated frequency responses at tested frequencies
4 load('H.mat'); % H is a 1xL matrix with the estimated freq resp
5 load('Omega.mat'); % Omega is a 1xL matrix with the corresponding freq
6
7 % number of frequencies tested
8 L=length(Omega);
9
10 % assert that enough frequencies were tested
11 assert(...); % assert(cond) throws an error if cond is false.
12
13 % matrices for least-squares problem
14 F = zeros(2*L, 3);
15 G = zeros(2*L, 1);
16
17 for l=0:L-1
18 R = abs(H(l+1)); % the magnitude of the estimated freq resp
19 Phi = phase(H(l+1)); % the phase of the estimated freq resp
20 Om = Omega(l+1); % the freq corresponding to the estimated freq resp
21
22 % populate F and G
23 F(2*l+1,:) = ...;
24 F(2*l+2,:) = [-R*sin(Phi-Om), 0, -sin(Om)];
25 G(2*l+1) = R*cos(Phi);
26 G(2*l+2) = R*sin(Phi);
27 end
28
29 % now estimate the transfer function coefficients using least-squares
30 Theta = F\G;
31
32 % define the transfer function;
33 % remember, a0 = 1 is not in the Theta parameter vector
34 a = [1; Theta(1)].';
35 b = Theta(2:3).';
```

Question 47: Which option correctly completes line 11?
Options:
(1) assert(2*L>6);
(2) assert(2*L>5);
(3) assert(2*L>3);
(4) assert(2*L>2);
(5) assert(2*L>1);
Award full credit only for unambiguous selection of option (2): assert(2*L>5);. Otherwise 0 unless the rubric allows partial credit.

q48 - The objective of the following MATLAB code is to estimate the transfe... - 1 point:

Question (q48): The objective of the following MATLAB code is to estimate the transfer function coefficients a_1, b_0 and b_1 of H(z) = (b_0 + b_1 z^-1) / (1 + a_1 z^-1), given estimated frequency responses H_hat(Omega) for frequencies INCLUDING Omega = 0 and Omega = pi, saved in MATLAB vectors H.mat and Omega.mat, respectively.

```
1 %% First order transfer function estimation
2
3 % load estimated frequency responses at tested frequencies
4 load('H.mat'); % H is a 1xL matrix with the estimated freq resp
5 load('Omega.mat'); % Omega is a 1xL matrix with the corresponding freq
6
```

```
7 % number of frequencies tested
8 L=length(Omega);
9
10 % assert that enough frequencies were tested
11 assert(...); % assert(cond) throws an error if cond is false.
12
13 % matrices for least-squares problem
14 F = zeros(2*L, 3);
15 G = zeros(2*L, 1);
16
17 for l=0:L-1
18 R = abs(H(l+1)); % the magnitude of the estimated freq resp
19 Phi = phase(H(l+1)); % the phase of the estimated freq resp
20 Om = Omega(l+1); % the freq corresponding to the estimated freq resp
21
22 % populate F and G
23 F(2*l+1,:) = ...;
24 F(2*l+2,:) = [-R*sin(Phi-Om), 0, -sin(Om)];
25 G(2*l+1) = R*cos(Phi);
26 G(2*l+2) = R*sin(Phi);
27 end
28
29 % now estimate the transfer function coefficients using least-squares
30 Theta = F\G;
31
32 % define the transfer function;
33 % remember, a0 = 1 is not in the Theta parameter vector
34 a = [1; Theta(1)].';
35 b = Theta(2:3).';
```

Question 48: Which option correctly completes line 23?
Options:
(1) F(2*l+1,:) = [-R*cos(Om), 1, cos(Phi-Om)];
(2) F(2*l+1,:) = [-R*cos(Phi-Om), 1, cos(Phi-Om)];
(3) F(2*l+1,:) = [R*cos(Phi-Om), 1, cos(Om)];
(4) F(2*l+1,:) = [-R*cos(Phi-Om), 0, cos(Om)];
(5) F(2*l+1,:) = [-R*cos(Phi-Om), 1, cos(Om)];
Award full credit only for unambiguous selection of option (5): F(2*l+1,:) = [-R*cos(Phi-Om), 1, cos(Om)];. Otherwise 0 unless the rubric allows partial credit.

q49 - Question 49: To determine the frequency response H(Omega) of a stable... - 1 point:

Question (q49): Question 49: To determine the frequency response H(Omega) of a stable LTI system at frequency Omega = pi/4, a sinusoidal input {u_e[n]} = {A cos(pi/4 n)} is applied and the output {y_m[n]} is measured. The first N_T samples are discarded and the following N samples are used to calculate the DFT coefficients Y_m[l] = sum_{n=N_T}^{N_T+N-1} y_m[n] e^(-j 2pi/N l n) and U_e[l] = sum_{n=N_T}^{N_T+N-1} u_e[n] e^(-j 2pi/N l n), where N_T = 200 and N = 1000. Which is the frequency response estimate at Omega = pi/4?
Options:
(1) H_hat(pi/4) = Y_m[150] / U_e[150]
(2) H_hat(pi/4) = Y_m[25] / U_e[25]
(3) H_hat(pi/4) = Y_m[250] / U_e[250]
(4) H_hat(pi/4) = Y_m[4] / U_e[4]
(5) H_hat(pi/4) = Y_m[50] / U_e[50]
(6) H_hat(pi/4) = Y_m[125] / U_e[125]
Award full credit only for unambiguous selection of option (6): H_hat(pi/4) = Y_m[125] / U_e[125]. Otherwise 0 unless the rubric allows partial credit.

q50 - Question 50: You conducted three experiments to identify an unknown t... - 1 point:

Question (q50): Question 50: You conducted three experiments to identify an unknown transfer function and obtained the magnitude response estimates |H_hat(Omega = 0)| = 0.05, |H_hat(Omega = pi/2)| = 0.39, |H_hat(Omega = pi)| = 1.96. Which

of the following transfer functions fits your experiments best?
Options:
(1) H(z) = 1/2 (z + 1) / (z - 0.5)
(2) H(z) = 1/2 (z^2 - 0.5) / (z^2 + 0.5)
(3) H(z) = 1/3 z / (z + 0.75)
(4) H(z) = 1/3 (z - 1) / (z + 2/3)
Award full credit only for unambiguous selection of option (4): H(z) = 1/3 (z - 1) / (z + 2/3). Otherwise 0 unless the rubric allows partial credit.